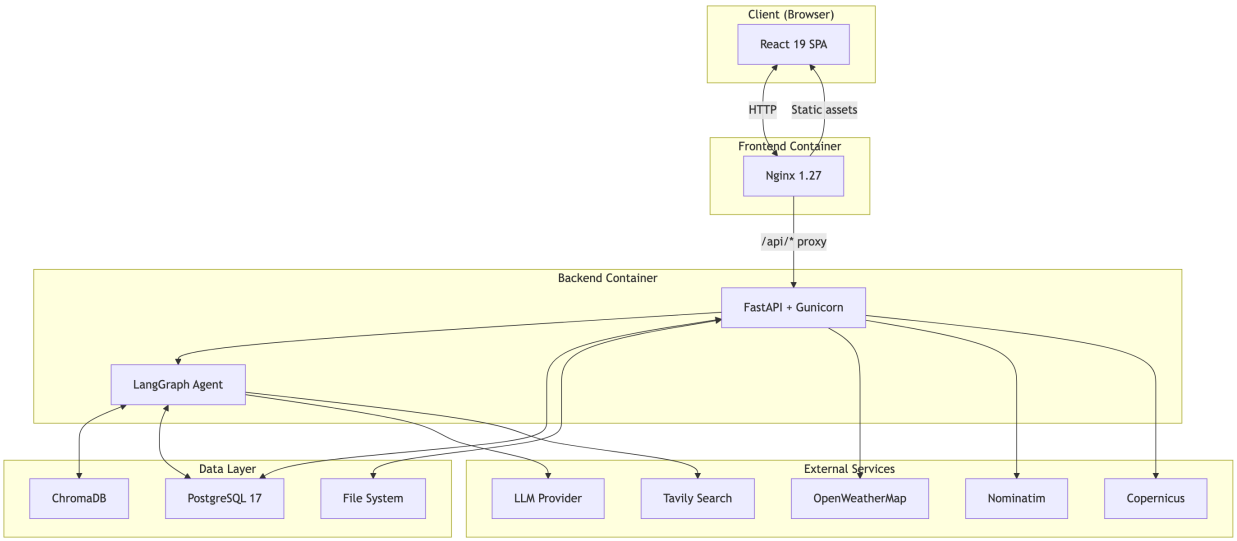
System Design

1. System Architecture

1.1 Architecture Diagram



1.2 Component Descriptions

Component	Technology	Responsibility
Frontend SPA	React 19, Vite	Single-page application providing the chat interface, dashboard, interactive map, and drone/satellite imagery panels.
Reverse Proxy	Nginx 1.27 Alpine	Serves static frontend assets and proxies `/api/` requests to the backend. Handles SPA routing via `try_files $uri /index.html`.
API Server	FastAPI, Gunicorn (2 Uvicorn workers, 120s timeout)	REST API layer handling authentication, file management, farm data CRUD, and proxying third-party API calls. All external API calls are made server-side.
AI Agent	LangGraph, LangChain	RAG-powered conversational agent with three tools: knowledge base retrieval, weather data, and optional internet search. Parses structured task and chart blocks from LLM output.
Vector Store	ChromaDB	Stores document embeddings for similarity search. Pre-built index baked into the Docker image; user uploads indexed at runtime with per-user metadata scoping.

Component	Technology	Responsibility
Database	PostgreSQL 17	Stores user accounts, farm data (with JSONB for polygon nodes and weather), and LangGraph conversation checkpoints. Accessed via an async connection pool (min 2, max 10).
File System	Docker volumes	Stores user-uploaded documents (`data/uploads/{user_id}/`) and computed drone vegetation index images (`data/ndvi_images/`).
LLM Provider	Azure OpenAI (default), OpenAI, Google GenAI, Anthropic, Ollama	Provides chat model inference and text embeddings. Configurable via `MODEL_PROVIDER` and `MODEL_NAME` environment variables.
OpenWeatherMap	REST API	Supplies current weather conditions and 5-day forecasts for farm coordinates.
Nominatim	REST API (OpenStreetMap)	Forward geocoding (address to coordinates) and reverse geocoding (coordinates to place name).
Copernicus Sentinel Hub	REST API	Fetches Sentinel-2 satellite imagery for server-side NDVI computation.
Tavily Search	LangChain tool	Real-time internet search, conditionally included in the agent's tool set based on user toggle.

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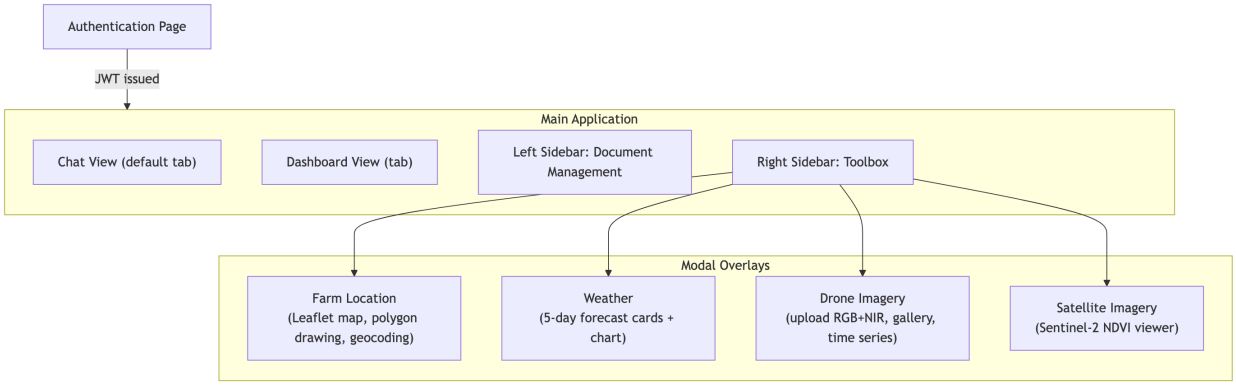
2. Site Map



Authentication Page — Login and registration forms. On successful authentication, a JWT is stored in `localStorage` and the user is redirected to the main application.

Chat View — Default view. Displays the message history with markdown rendering, inline Recharts charts, colour-coded task cards, and source citations. The input area includes a text field and an internet search toggle.

Dashboard View — Accessible via a tab. Aggregates 5-day weather forecast, seasonal context, active tasks from chat responses, and a field health NDVI chart from drone imagery history.

Left Sidebar — Document management panel. Supports file upload (click or drag-and-drop) for `.md`, `.pdf`, `.txt`, `.csv`, and `.json` files. Lists uploaded files with type icons and deletion controls.

Right Sidebar — Toolbox with buttons for Farm Location, Weather, Drone Monitoring, Satellite Imagery, and Web Search toggle. Each button opens its corresponding modal overlay.

Modal Overlays — Full-screen overlays for interactive tools: farm polygon drawing on a satellite map, weather forecast display, drone image upload and gallery, and satellite NDVI viewer.

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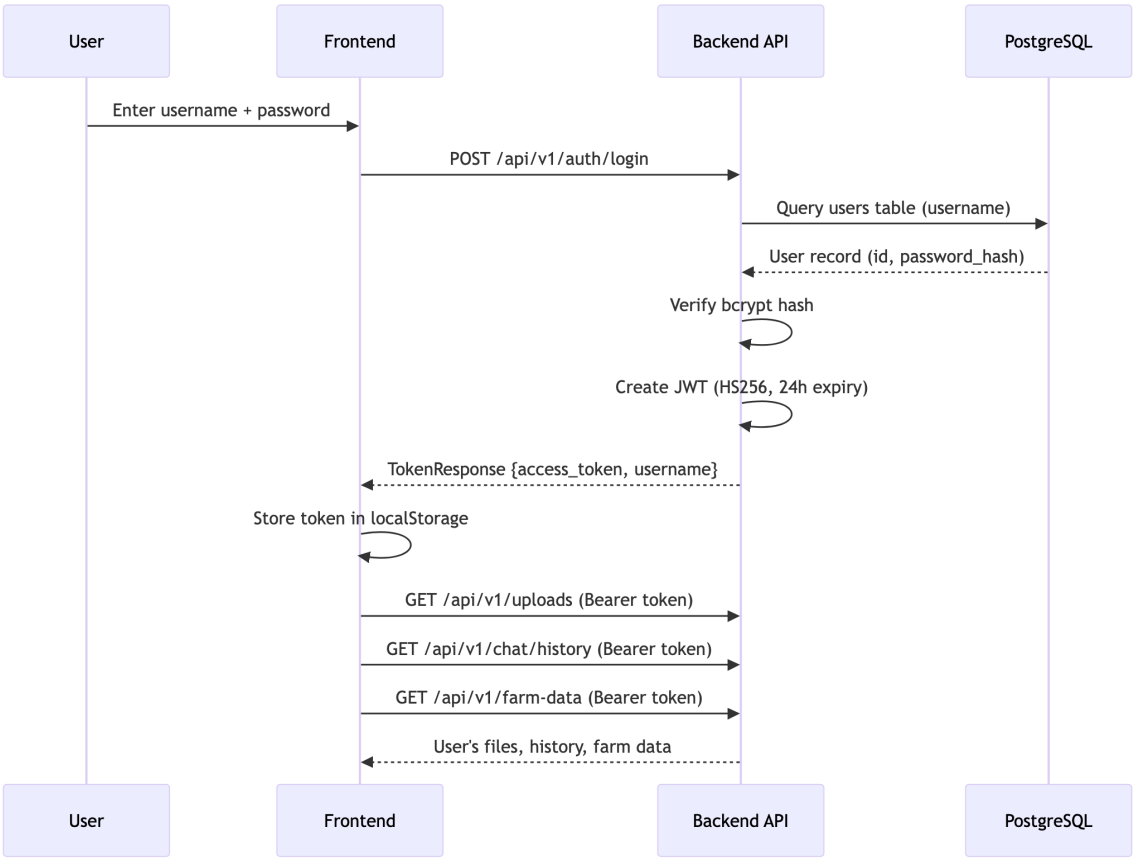
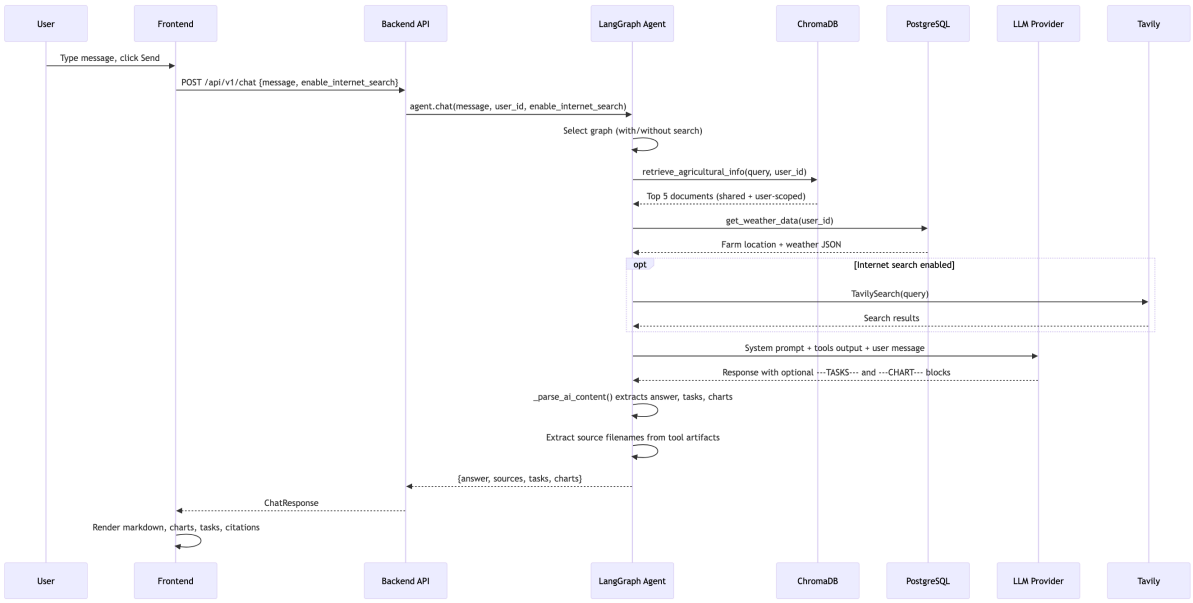
3. Sequence Diagrams

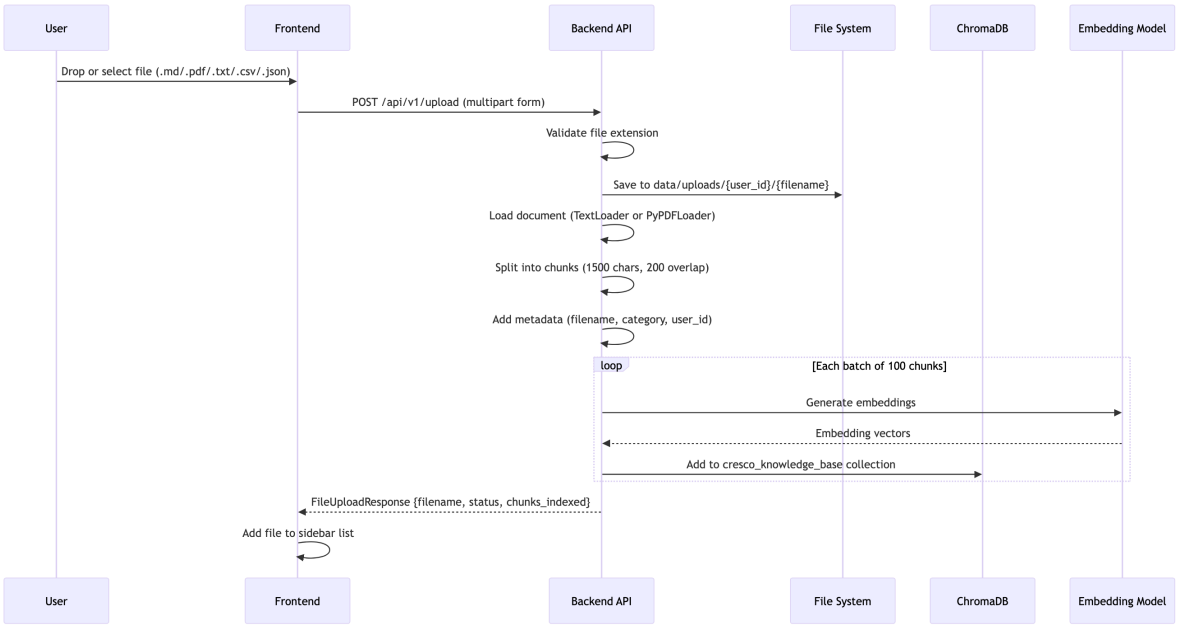
3.1 User Authentication



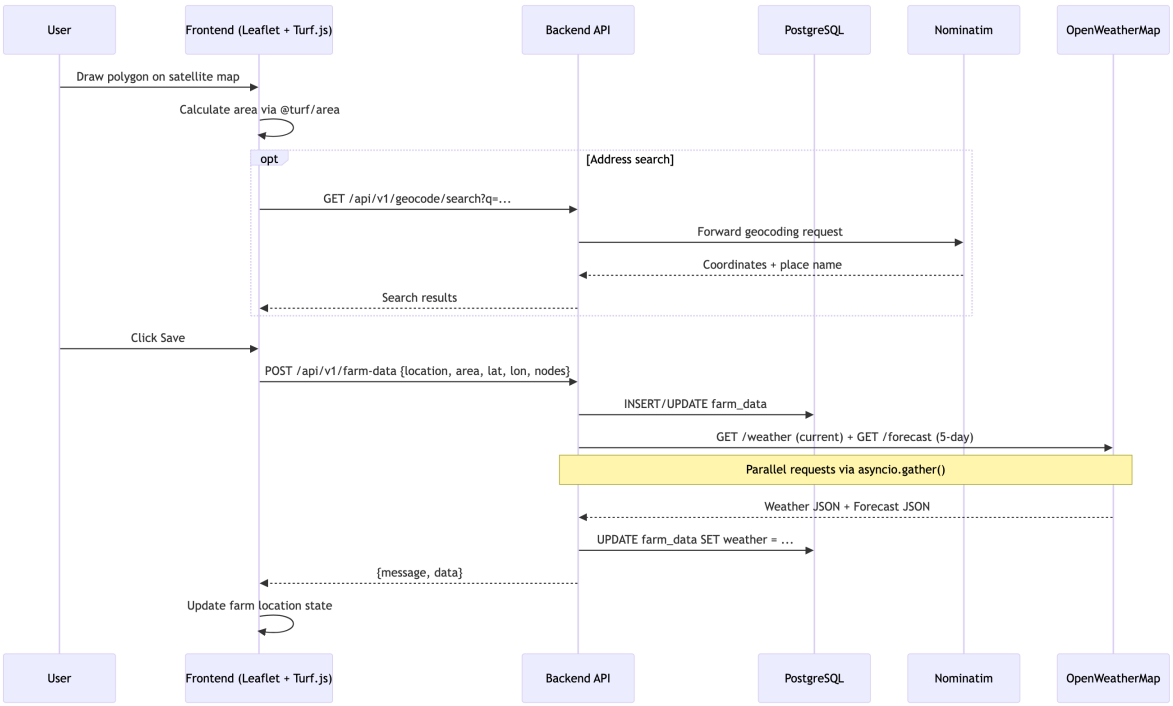
3.2 Chat with RAG



3.3 File Upload and Indexing



3.4 Farm Location and Weather



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4. Design Patterns

4.1 Singleton

Two singleton variants are used to ensure expensive resources are initialised once per process.

LRU cache singleton (`config.py`): The `get_settings()` function uses `@lru_cache` to create a single `Settings` instance. Tests reset it via `get_settings.cache_clear()`.

Module-level singleton (`agent.py`, `embeddings.py`, `retriever.py`): A module-level `_variable = None` is lazily initialised by a `get_*()` function. This pattern is used for the `CrescoAgent`, `AzureOpenAIEmbeddings`, `Chroma` vector store, and retriever instances. Tests reset them by setting the module variable to `None`.

4.2 Factory

`create_app()` in `main.py` is a factory function that constructs and returns a fully configured FastAPI application with CORS middleware, routers, and a lifespan manager. This separates application construction from the module-level `app` instance, enabling test clients to create isolated app instances.

4.3 Strategy

The `CrescoAgent` pre-builds two LangGraph agents at initialisation: `_agent_with_search` (includes Tavily) and `_agent_no_search` (excludes Tavily, appends a "search disabled" addendum

to the prompt). The `chat()` method selects the appropriate agent at runtime based on the user's `enable_internet_search` flag, avoiding the overhead of rebuilding the graph on every request.

4.4 Dependency Injection

FastAPI's `Depends()` mechanism injects shared resources into route handlers:

- `get_db_pool(request)` — extracts the async connection pool from `app.state`
- `get_agent_dep(request)` — passes the checkpointer from `app.state` to `get_agent()`
- `get_current_user(credentials)` — validates the JWT Bearer token and returns user identity
- `get_current_admin(current_user)` — chains on `get_current_user` and enforces admin role
- `get_settings()` — injects the cached settings instance

This allows routes to declare their dependencies declaratively and enables tests to override them via `app.dependency_overrides`.

4.5 Proxy

All third-party API calls (OpenWeatherMap, Nominatim, Copernicus) are proxied through backend endpoints using `httpx.AsyncClient`. The frontend never contacts external services directly. This keeps API keys server-side (NFR-02) and provides a single point for timeout enforcement, error handling, and response transformation.

4.6 Repository

The `db.py` module provides a functional repository interface over PostgreSQL. Async functions (`save_farm_data`, `get_farm_data`, `update_farm_weather`) accept a connection pool as the first argument and encapsulate all SQL queries. A separate sync function (`get_farm_data_sync`) serves the agent tool, which runs in a LangGraph thread pool where async calls are not available.

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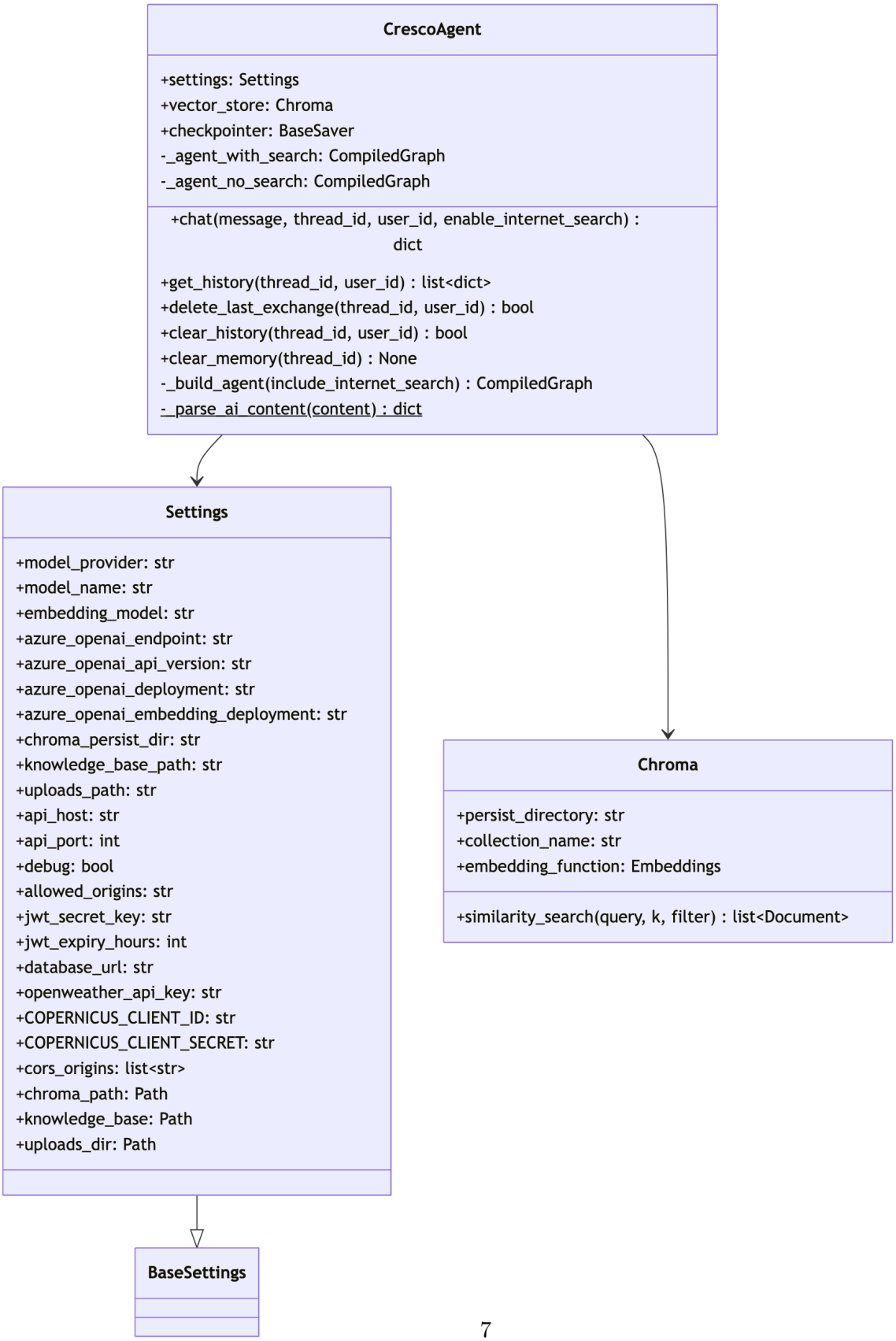
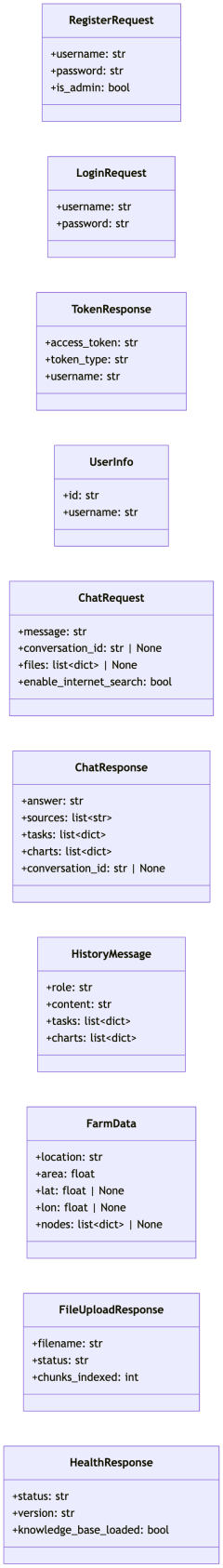
5. Class Diagrams

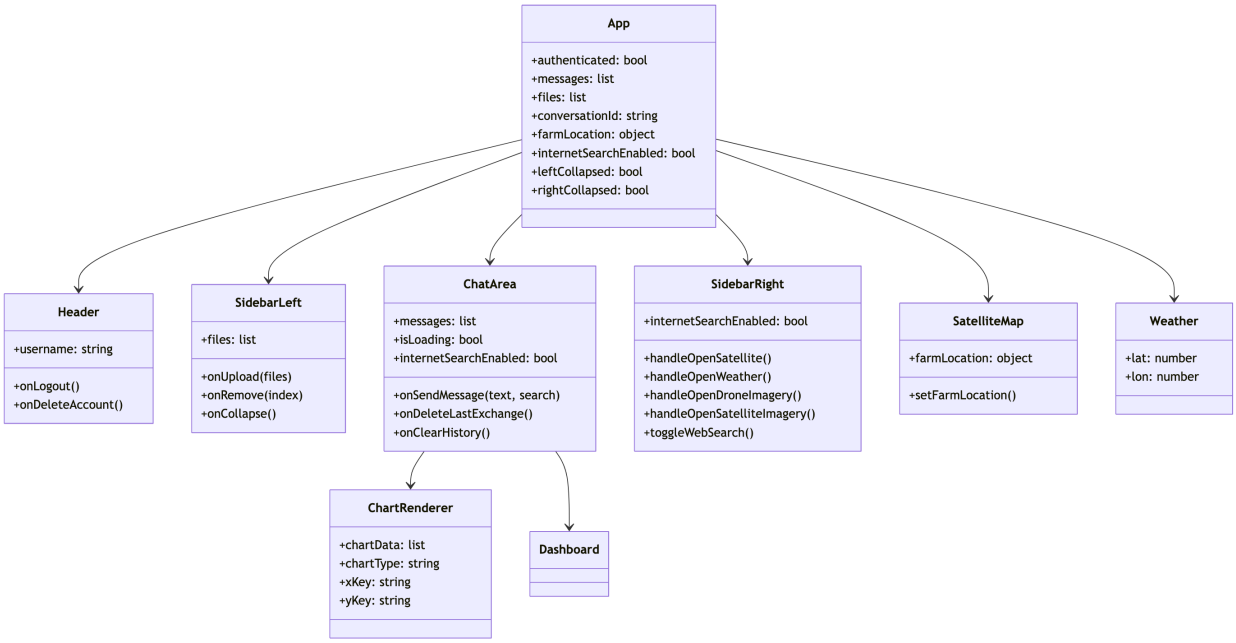
5.1 Backend Classes



5.2 Pydantic Request/Response Models



5.3 Frontend Component Hierarchy



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6. Data Storage

6.1 PostgreSQL

PostgreSQL 17 stores relational data across two application tables and auto-generated LangGraph checkpointer tables.

Table: `users`

```
CREATE TABLE users (
    id            TEXT PRIMARY KEY,
    username      TEXT UNIQUE NOT NULL,
    password_hash TEXT NOT NULL,
    is_admin      BOOLEAN NOT NULL DEFAULT FALSE,
    created_at    TIMESTAMPTZ NOT NULL
);
```

Passwords are hashed with bcrypt before storage. The `id` field is a UUID generated at registration time.

Table: `farm_data`

```
CREATE TABLE farm_data (
    user_id  TEXT PRIMARY KEY,
    location TEXT,
    area     DOUBLE PRECISION,
    lat      DOUBLE PRECISION,
    lon      DOUBLE PRECISION,
```

```
    nodes    JSONB,
    weather  JSONB
);
```

The `nodes` column stores the farm polygon boundary as a JSONB array of `{lat, lng}` objects. The `weather` column caches the most recent OpenWeatherMap response (current conditions and 5-day forecast) to make it available to the agent's weather tool without an API call.

LangGraph Checkpointer Tables — Auto-created by `AsyncPostgresSaver.setup()`. Stores serialised conversation state (all messages, tool calls, and tool responses) keyed by `thread_id`, enabling conversation persistence across server restarts.

6.2 ChromaDB

ChromaDB provides the vector store for Retrieval-Augmented Generation.

Property	Value
Collection	`cresco_knowledge_base`
Embedding model	Azure OpenAI `text-embedding-ada-002` (configurable)
Chunk size	1500 characters, 200 character overlap
Splitter	`RecursiveCharacterTextSplitter` with separators: section breaks, headers, paragraphs, lines, words
Search type	MMR (Maximum Marginal Relevance), k=5, fetch_k=10
Metadata per chunk	`filename`, `category`, `user_id`, `chunk_index`

Document scoping uses the `user_id` metadata field. Shared knowledge base documents are tagged with `user_id = "__shared__"`. User uploads are tagged with the user's UUID. The retrieval tool filters with `$or: [user_id == "__shared__", user_id == current_user]` to return both shared and user-specific documents.

6.3 File System

Path	Contents	Lifecycle
`data/knowledge_base/`	Shared agricultural knowledge base documents (.md, .pdf, .txt)	Baked into Docker image at build time
`data/chroma_db/`	Pre-built ChromaDB index over the knowledge base	Baked into Docker image at build time
`data/uploads/{user_id}/`	User uploaded documents	Created at runtime; deleted on file removal or account deletion

Path	Contents	Lifecycle
`data/ndvi_images/`	Computed drone vegetation index images with JSON metadata	Created at runtime; deleted on image removal or account deletion

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7. Packages and APIs

7.1 Key Packages

Backend (Python 3.12):

Package	Version	Purpose
fastapi	>= 0.115	Async REST API framework
gunicorn + uvicorn	>= 22.0, >= 0.32	Production ASGI server (2 Uvicorn workers, 120s timeout)
langchain + langgraph	>= 0.3, >= 0.2	LLM orchestration and graph-based agent with tool use
langchain-openai	>= 0.2	Azure OpenAI and OpenAI chat model and embedding integration
langchain-chroma + chromadb	>= 0.1, >= 0.5	Vector store for RAG document retrieval
langchain-tavily	>= 0.2	Internet search tool for the agent
psycopg[binary,pool]	>= 3.2	Async PostgreSQL driver with connection pooling
langgraph-checkpoint-postgres	>= 2.0	LangGraph conversation state persistence in PostgreSQL
pyjwt[crypto] + bcrypt	>= 2.8, >= 4.1	JWT token creation/validation and password hashing
httpx	>= 0.27	Async HTTP client for proxying third-party APIs
rasterio + pillow + matplotlib	>= 1.3, >= 10.0, >= 3.0	Drone and satellite image processing
pydantic-settings	>= 2.0	Environment variable configuration via `.env`
pypdf	>= 4.0	PDF document loading for the knowledge base

Frontend (Node 22):

Package	Version	Purpose
react + react-dom	19.2	UI framework
vite	7.2	Build tool and development server
leaflet + react-leaflet	1.9, 5.0	Interactive satellite map with polygon drawing
recharts	3.7	Chart rendering (bar, line, pie) inline in chat
react-markdown	10.1	Markdown rendering for AI responses
remark-gfm + remark-math	4.0, 6.0	GitHub Flavoured Markdown and math notation plugins
rehype-katex + katex	7.0, 0.16	LaTeX math typesetting
@turf/area + @turf/helpers	—	GeoJSON polygon area calculation
lucide-react	0.562	Icon library
vitest + @testing-library/react	4.0, 16.3	Unit testing framework

7.2 API Endpoints

All endpoints are served under `/api/v1`. Endpoints marked with a lock require a valid JWT Bearer token.

Authentication

Method	Path	Description	Auth
POST	`/auth/register`	Create account, returns JWT	No
POST	`/auth/login`	Authenticate, returns JWT	No
DELETE	`/auth/me`	Delete own account	Yes

Chat

Method	Path	Description	Auth
POST	`/chat`	Send message, returns AI response with tasks/charts/sources	Yes
GET	`/chat/history`	Fetch full conversation history	Yes
DELETE	`/chat/history`	Clear all conversation history	Yes
DELETE	`/chat/last-exchange`	Remove the last question-answer pair	Yes

Farm Data

Method	Path	Description	Auth
POST	`/farm-data`	Save farm location, polygon, and area	Yes
GET	`/farm-data`	Retrieve saved farm data	Yes

Weather and Geocoding

Method	Path	Description	Auth
GET	`/weather?lat=&lon=`	Fetch current weather and 5-day forecast (proxied from OpenWeatherMap)	Yes
GET	`/geocode/search?q=`	Forward geocoding (proxied from Nominatim)	Yes
GET	`/geocode/reverse?lat=&lon=`	Reverse geocoding (proxied from Nominatim)	Yes

File Management

Method	Path	Description	Auth
POST	`/upload`	Upload and auto-index a document	Yes
GET	`/uploads`	List uploaded files	Yes
DELETE	`/upload/{filename}`	Delete file and its ChromaDB chunks	Yes

Drone Imagery

Method	Path	Description	Auth
POST	`/droneimage?index_type=`	Upload RGB + NIR images, compute vegetation index	Yes
GET	`/images`	List saved drone analysis images	Yes
GET	`/images/{filename}`	Retrieve a specific drone image	Yes
DELETE	`/images/{filename}`	Delete a drone image	Yes
PATCH	`/images/{filename}/timestamp`	Edit capture timestamp	Yes

Satellite and System

Method	Path	Description	Auth
POST	`/satellite-image`	Fetch Sentinel-2 NDVI for farm coordinates	Yes
POST	`/index`	Re-index the knowledge base	Yes
GET	`/health`	Health check	No
DELETE	`/account`	Delete user account and all associated data	Yes

7.3 External API Integrations

API	Purpose	Backend Access Point	Auth Method
OpenWeatherMap	Current weather and 5-day forecast	`GET /weather` proxied via `httpx`	API key (server-side)
Nominatim (OpenStreetMap)	Forward and reverse geocoding	`GET /geocode/search`, `GET /geocode/reverse` proxied via `httpx`	None (User-Agent header)
Copernicus Sentinel Hub	Sentinel-2 satellite imagery for NDVI	`POST /satellite-image` via `satellite_image.py`	OAuth2 client credentials
Tavily Search	Real-time internet search within agent	Invoked as a LangGraph tool (no direct endpoint)	API key (server-side)
Azure OpenAI	Chat model inference and text embeddings	Invoked within `CrescoAgent` and `get_embeddings()`	API key (server-side)
OpenAI / Google GenAI / Anthropic / Ollama	Alternative LLM providers	Invoked within `CrescoAgent` via `init_chat_model()`	Provider-specific API key (server-side)